

Alec Mackay

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PROFILE

Second-year BA Product and Industrial Design student at Central Saint Martins (UAL), with experience developing sustainable product concepts from research and sketching through CAD, rendering and physical prototyping. Hands-on experience across Fusion 360, SolidWorks and Blender, with technical drawing, materials experimentation and wood/metal fabrication. Interested in consumer-centred design, sustainable materials and packaging development, and seeking a 12-month industrial placement to contribute to a cross-functional design team.

EDUCATION

BA Product & Industrial Design — Central Saint Martins, UAL, London 09/2025 – Present
Currently in 2nd year of study (2026/27)

UAL Foundation Diploma in Art & Design — University of the Arts London 09/2024 – 05/2025

A-Levels — College of Richard Collyer, Horsham 09/2022 – 06/2024
English Literature (B) • History (C) • Mathematics (D)

GCSEs — Dubai English Speaking College 2017 – 2022
11 GCSEs including English Literature, Business Studies, Media Studies, Spanish and Mathematics.

DESIGN PROJECTS

Radia — Collapsible Firepit — UAL Foundation Diploma 2025

- Developed a collapsible steel firepit from primary research and concept development through CAD, physical prototyping and final manufacture, focused on longevity, portability and material efficiency.
- Produced scale models and renders in SolidWorks and Blender before hand-cutting, sanding, polishing and bending steel to create the final prototype.
- Selected for the UAL Millbank Exhibition 2025.

Glowing Lily — CO₂ Sensor — CSM Project 2026

- Developed a screen-free, app-free CO₂ monitoring concept from secondary research into indoor air quality and CO₂ exposure, distilling findings into a simplified 400–2500 ppm product gauge.
- Designed the product, component layout and housing in Fusion 360 and Blender, using 1:1 scale component and housing studies to refine the form.
- Produced a technical package covering components, materials, dimensions and manufacture; concept includes PLA/TPU 3D-printed parts.

Living Habitat — Rewilding Birdhouse — CSM Project 2025

- Designed a multi-species birdhouse exploring sustainable and rewilding-led design, incorporating the foundations for a living roof.
- Developed the design from hand-drawn technical drawings through paper and cardboard models to a full-scale wooden prototype using workshop machinery.

Turkish Spice Pot Concept — UAL Foundation Diploma 2024

- Developed a multi-piece spice dish through historical, modern and chemical research into peppercorns, linking the concept to Turkish cultural references and geometric form.
- Analogue concept development using iterative sketching and physical form exploration.

DESIGN & TECHNICAL SKILLS

CAD & Visualisation: Fusion 360, SolidWorks, Blender, CAD modelling, technical drawing, rendering

Design Development: Research, concept generation, sketching, physical prototyping, materials experimentation, product development

Manufacturing: Wood and metal fabrication, workshop machinery, digital manufacturing, 3D-printed prototype components

Sustainability: Sustainable and circular design principles, materials research, rewilding-led design

Consumer & Digital: User-centred design, consumer insight gathering, UX fundamentals, AI tools and workflows

Additional: Teamwork, communication, project/time management, full UK driving licence

RELEVANT COURSEWORK & CPD

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- SolidWorks 3D CAD for Education Specialisation** — Coursera / Dassault Systèmes *(in progress)*
 - Use Fusion 360 to Turn Your Ideas into Designs** — LinkedIn Learning *(in progress)*
 - Digital Manufacturing: Introduction and Smart Design** — Coursera / L&T EduTech *(in progress)*
 - Google UX Design Professional Certificate** — Coursera / Google *(in progress)*
 - AI Fluency curriculum** — Claude Academy / Anthropic *(in progress)*

WORK EXPERIENCE

Bartender — The Parrot Inn, Forest Green 07/2025 – Present

- Front-of-house service, stock and equipment care in a busy pub environment.

Bartender — The Red Lion, Guildford 05/2025 – 07/2025

- Full bar service and customer-facing stock advice.

Bartender — Cranleigh Rugby Football Club 09/2024 – Present

- Front-of-house service, opening/closing procedures and equipment care alongside full-time university study and university rugby.